

SIMRAN ABHAY SINHA

sinha.sim@northeastern.edu • +1 (857) 381-5028 • Boston, MA • [LinkedIn](#) • [GitHub](#) • [Tableau](#) • [Portfolio](#)

SUMMARY

Data Analytics Engineering graduate student at Northeastern University with hands-on experience building end-to-end analytics and AI systems. Proven track record delivering business impact through machine learning, dashboards, and data pipelines across 10+ global projects. Strong focus on translating data into decisions, with experience in LLM applications, recommendation systems, and scalable analytics infrastructure.

EDUCATION

Northeastern University, Boston, USA <i>Master of Science in Data Analytics Engineering</i> Coursework: Applied Generative AI, Deep Learning for AI, Data Management Analytics, Structured Data Analytics, Storytelling with Data	May 2026 GPA 3.8/4.0
K. J. Somaiya Institute of Technology, Mumbai, India <i>Bachelor of Technology in Electronics and Telecommunication</i> Coursework: Neural Networks and Deep Learning, Machine Learning, Big Data Analytics, Project Management, Data Mining	June 2023 GPA 3.7/4.0

SKILLS

- **Programming Languages:** SQL, Python, CSS, HTML, R programming
- **Machine Learning / AI:** Scikit-learn, NLP, Recommendation Systems, LLMs, Prompt Engineering
- **Data & Visualization:** Power BI, Tableau, Excel, DAX
- **Cloud & Tools:** AWS (SageMaker, RDS, S3), Streamlit, Git, VS Code
- **Concepts:** ETL, Data Modeling, Predictive Analytics, Experimentation

WORK EXPERIENCE

Data Science Lead - Crewasis, New York, USA	Sept 2025 - Dec 2025
• Increased marketing ROI by 18% by analyzing consumer data across 10+ enterprise projects using Python, SQL, SageMaker, and Bedrock • Reduced model deployment time by 35% by building scalable ML pipelines on AWS (SageMaker, RDS, Lambda) improving production reliability • Increased stakeholder reporting efficiency by 40% by architecting Power BI dashboards and maintaining the Crewasis DS Portal • Improved personalization by developing NLP-based recommendation engine on 200K+ consumer records, cutting deployment time by 20% • Accelerated team output by mentoring 9+ junior data scientists on modeling and stakeholder communication, aligning with executive priorities	
Data Science Intern - Prodigy, Mumbai, India	Jul 2024 - Aug 2024
• Increased marketing ROI by 25% by fine-tuning Scikit-Learn classification models to predict customer purchase behavior across large datasets • Improved campaign engagement by 20% by building Power BI dashboards for customer segmentation, KPI tracking, and performance visibility • Reduced manual processing time by 30% by automating end-to-end ETL pipelines in Python, improving data accuracy across workflows • Cut onboarding time by 20% by documenting scalable ETL workflows and modeling standards, enabling reproducible analytics across 3+ teams	
Data Science Intern- Champion Learns, Mumbai, India	Jan 2023 - Dec 2023
• Boosted qualified lead conversion by 25% by building precision targeting models using Python and SQL to analyze customer behavior • Improved campaign personalization by 30% by applying K-Means clustering to segment 10K+ customers by behavioral and engagement metrics • Reduced reporting turnaround by 30% by automating data cleaning and reporting workflows, eliminating 15+ hours of manual work weekly • Shaped marketing strategy by presenting data-backed recommendations to 5+ stakeholders, influencing channel allocation and targeting	

PROJECTS

AI Budget Travel Scout – LLM-Powered Travel Planner Python (LLMs, APIs), Streamlit, Semantic Search, Prompt Engineering	[Link]
• Built an LLM-powered budget travel planner, automating itinerary generation and reducing manual trip planning time by ~40% • Improved recommendation relevance by 35% by implementing semantic search and constraint filtering across budget, location, and duration • Developed an interactive Streamlit app with real-time APIs, enabling instant itinerary generation with 100% budget adherence	
VisionScript (BLIP Image Captioning) Python (PyTorch, Transformers), BLIP, Gradio, Hugging Face Spaces	[Link]
• Built a full-stack multimodal AI app (BLIP + Gradio) for real-time image captioning and visual Q&A, achieving ~1–2s inference latency • Designed user-controlled caption features (detail level, temperature) enabling 3+ output modes to balance accuracy and creativity • Deployed on Hugging Face Spaces with GitHub CI/CD, achieving 100% cloud accessibility and a fully reproducible end-to-end pipeline latency	
SpotIt – Object Detection System Python, YOLOv8, PyTorch, PIL, Google Colab	[Link]
• Built an end-to-end custom object detection pipeline using YOLOv8, progressing from single-object classification to multi-object localization • Generated 100+ synthetic multi-object images using PIL with IoU-based overlap control and auto-generated YOLO-format annotations • Trained custom YOLOv8 model achieving stable mAP, precision, and recall across validation set with augmentation-driven performance gains	
Vital Sign Manager (Hospital Management System) SQL, Python, NOSQL (MongoDB)	[Link]
• Reduced patient data retrieval time by 40% by building a hybrid SQL/NoSQL hospital management database for 100+ patient records • Improved anomaly detection by building Python visualizations for critical vital signs, reducing clinical response time by 25% • Strengthened data privacy by integrating role-based access controls, ensuring HIPAA compliance across 100+ patient records	
Predicting Diabetes Risk using Health Indicator Python, Scikit-Learn, Logistic Regression, Random Forest, XGBoost	[Link]
• Built and evaluated Logistic Regression, Random Forest, and XGBoost to predict diabetes risk, improving early diagnosis accuracy by 18% • Boosted model interpretability by engineering 10+ features from health and demographic data (blood pressure) improving clinical relevance • Delivered end-to-end ML pipeline with preprocessing, tuning, and evaluation, enabling healthcare teams to prioritize 3x faster intervention	